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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Mini Project Phase-01
Presentation on

"AI-BASED RESTAURANT DEMAND FORECASTING AND REAL-TIME FOOD WASTE MANAGEMENT SYSTEM"

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PRESENTATION OVERVIEW

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A systematic walk-through from problem definition to technical architecture and implementation tools.

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Restaurant operational context, preparation dilemmas, and project concept

02 **Problem Statement**

Core issues of demand uncertainty, limited visibility, and unmanaged surplus

03 **Objectives**

Five targeted technical goals and realistic expected project outcomes

04 **Existing System and Research Gap**

Comparative analysis of current approaches and unaddressed system needs

05 **Proposed System**

Integrated end-to-end operational workflow and key system advantages

06 **System Architecture / Block Diagram**

Five-layer architectural design, data pipelines, and periodic feedback loop

07 **Software Requirements / Tools Used**

Development software stack, machine learning libraries, and hardware specifications

Introduction

- Restaurants must estimate the quantity of each food item to prepare before actual customer demand is fully known.
- Excess preparation can cause food waste, higher operating costs, and inefficient ingredient utilization.
- Insufficient preparation can cause shortages, lost sales, and customer dissatisfaction.
- Demand fluctuates with weekdays, festivals, special occasions, group bookings, and unexpected orders.
- Machine learning and real-time order monitoring can support better demand and inventory decisions.

PROJECT IDEA

A web-based application that predicts restaurant food demand, monitors incoming orders, tracks inventory, and supports responsible redistribution of eligible surplus food.



Problem Statement

> “Restaurants often struggle to accurately predict customer demand, resulting in excess food preparation, food wastage, increased costs, and shortages of popular menu items. Traditional estimation methods based on manual judgment may not respond effectively to changing demand, special events, or unexpected customer orders. An intelligent system is needed to forecast demand using historical sales data, monitor orders in real time, support inventory decisions, and facilitate the redistribution of eligible surplus food.”



1. Demand Uncertainty

Actual customer demand can differ from expected demand.



2. Limited Real-Time Visibility

Manual tracking makes it difficult to respond promptly to changing sales and inventory levels.



3. Unmanaged Surplus

Leftover food may be wasted when safe redistribution is not coordinated.

Objectives

01

Predict Food Demand

Use historical sales and relevant contextual features to estimate demand for individual menu items.

02

Recommend Preparation Quantities

Help managers plan initial preparation using predicted demand and available inventory.

03

Monitor Orders in Real Time

Record customer orders and update sales and inventory as orders are entered.

04

Generate Demand and Inventory Alerts

Identify potential shortages or excess preparation when actual sales differ from expectations.

05

Manage Eligible Surplus Food

Record leftovers and facilitate collection coordination with participating food recovery organizations.

Expected Outcome: Better-informed preparation decisions, improved inventory utilization, and reduced avoidable food waste. No guaranteed reduction or perfect accuracy claims.

Existing System and Research Gap

Preliminary comparison of general existing approaches; no specific papers or statistics cited.

EXISTING APPROACH	METHOD	LIMITATION	RESEARCH GAP
Manual Demand Estimation	Staff experience and past observations	Subjective and may not adapt consistently to changing demand	Limited data-driven forecasting
Historical-Average Forecasting	Average sales from previous days or similar weekdays	May not capture complex demand patterns or unusual events	Need to evaluate more flexible forecasting models
Sales and Inventory Management Systems	Record orders, sales, and stock quantities	Basic systems may not provide integrated ML-based forecasting and dynamic recommendations	Connect demand estimates with live order and inventory updates
Surplus Food Donation Platforms	Connect surplus food providers with food recovery organizations	Primarily focus on redistribution rather than forecasting restaurant demand	Integrate demand planning with eligible surplus food coordination

Proposed System

A web-based application combining machine learning, live order monitoring, inventory tracking, and surplus food coordination.

1

Historical Data and Event Input

Historical sales, menu items, weekdays, bookings, known special events

2

AI-Based Demand Forecasting

A trained ML regression model estimates demand per menu item

3

Preparation Planning

Predictions and inventory guide initial preparation quantities

4

Real-Time Order Monitoring

Staff enter orders; sales and inventory records update

5

Dynamic Forecasting and Alerts

The existing trained model and recent order activity reassess remaining demand and flag potential shortages or excess

6

Surplus Food Management

Staff record leftovers and list eligible surplus for collection coordination with participating food recovery organizations

Historical Data → Demand Prediction → Preparation Planning → Live Order Tracking → Updated Forecasts and Inventory Alerts → Surplus Management

System Architecture / Block Diagram

LAYER 1: INPUT SOURCES

Historical sales dataset; menu items and prices; weekday and date information; festival, booking, and special-event inputs; customer orders entered by staff

LAYER 2: FRONTEND

HTML, CSS, JavaScript; manager dashboard; staff order-entry interface; event and inventory forms

LAYER 3: BACKEND APPLICATION (PYTHON FLASK)

User and role management; order processing; inventory updates; forecast requests; alert and report generation

LAYER 4A: MACHINE LEARNING MODULE

Python; pandas; scikit-learn; trained forecasting model; predicted demand and evaluation metrics

LAYER 4B: DATABASE MODULE (MYSQL)

Menu records; historical sales; customer orders; inventory; forecast records; waste and surplus collection records

LAYER 5: SYSTEM OUTPUTS

Predicted demand per item; suggested preparation quantities; updated sales and inventory; shortage or excess alerts; food waste reports; eligible surplus listings and collection requests

Periodic Feedback Loop: Recorded Actual Sales → Historical Dataset → Periodic Model Retraining (not after every order).

Software Requirements / Tools Used

Software Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python Flask
- **Machine Learning:** scikit-learn
- **Data Processing:** pandas, NumPy
- **Database:** MySQL
- **Development Tools:** Visual Studio Code, Jupyter Notebook
- **Operating System:** Windows or Linux

Hardware Requirements

- Laptop or desktop computer
- Recommended RAM 8 GB
- Modern multi-core processor
- Sufficient storage for application, dataset, and database
- Internet connection for package installation and optional hosted services

THANK YOU